Setdrift: Configuration Drift Detection and Repair for AI Coding-Agent Configuration

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Abstract

[PLACEHOLDER — Phase 10 (DELIV-04) fills in the real abstract once evidence-run results land. Source paragraph for the Setdrift/Robeyns name-collision differentiator: see `.planning/codebase/dissertation-a` — reuse that 145-word paragraph verbatim as the starting point, then extend with the real survive/null/die outcome once Phase 7 evidence lands.]

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1 Introduction

[PLACEHOLDER — Phase 10 (DELIV-04). Frames the falsifiable claim (design doc §1, `repo/sica-plugin/docs/design/2026-05-20-falsifia` and the observe→diagnose→patch→verify loop as the paper's contribution.]

2 Related Work

Setdrift shares its name with, but targets a fundamentally different layer of the system than, Robeyns et al.'s Self-Improving Coding Agent [1], which edits an agent's own source code to improve its own implementation. Setdrift instead optimizes AI-coding-agent *configuration* — skills, hooks, sub-agents, MCP wiring, CLAUDE.md — and treats source-code self-modification as an explicit anti-goal. [PLACEHOLDER — Phase 10 (DELIV-04) extends this section with the full five-axis differential (Chapter 2 of the dissertation) and the competitor-differential one-pager (WRITE-01).]

3 Methodology

[PLACEHOLDER — Phase 10 (DELIV-04). Skill-trigger macro-F1 as the primary metric, the three experiment arms (A/B/C), and the GEPA-based optimizer loop.]

4 Corpus Construction

[PLACEHOLDER — Phase 10 (DELIV-04). GitBug-Java-derived public corpus, the heuristic-mining precision gate, and inter-rater agreement methodology.]

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5 Results

[PLACEHOLDER — GATED ON REAL F1 (Phase 7 evidence runs, RUN-01..06). Phase 10 (DELIV-04) mechanically selects the matching survive/null/die branch template from `.planning/dissertation/06-results` once real numbers land — this section cannot be drafted before that gate opens.]

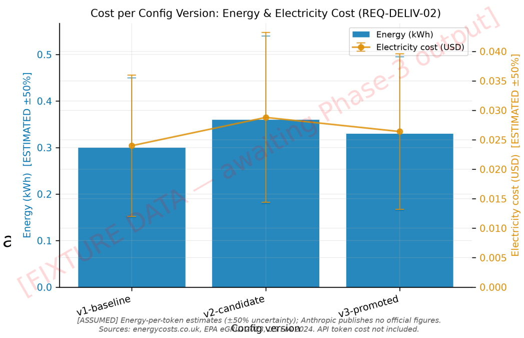

Figure 1: FIXTURE DATA — this figure is watermarked synthetic data used solely to verify the acmart `\includegraphics` + compile pipeline (WRITE-02 compile-smoke-test only, D-09). It is not dissertation content and will be replaced by Phase 9 (FIG-01) with a real, unwatermarked cost-delta figure once real experiment data lands.

6 Threats to Validity

[PLACEHOLDER — Phase 10 (DELIV-04). See `.planning/dissertation/07-thre` for the full-prose threats-to-validity chapter drafted in this phase (WRITE-03); this section is a compressed paper-length summary of that chapter.]

7 Conclusion

[PLACEHOLDER — Phase 10 (DELIV-04). Gated on the same survive/null/die decision rule (design doc §6) as the Results section.]

References

[1] Maxime Robeyns, Martin Szummer, and Laurence Aitchison. 2025. A Self-Improving Coding Agent. Live-verified 2026-07-27 via WebSearch (arxiv.org/pdf/2504.15228, huggingface.co/papers/2504.15228, and the paper's own GitHub repo github.com/MaximeRobeyns/self$_i$*mproving$_c$oding$_a$gent*)¯*arXivID, authorlist, and*2025−04−21*publicationdateconfirmedcorrect.PresentedattheICLR*2025*WorkshoponScalingS ImprovingFoundationModels.ThisistheRobeynsetal.*2025”*Self* − *ImprovingCodingAgent*”*referencecitedin.planning/codebase/dissertation*− *abstract* − *name* − *note.mdasthename* − *collisiondifferentiatorforthisdissertation′sChapter*2*five* − *axisdifferential..arXiv* : 2504.15228 [*cs.AI*]